

MACHINE LEARNING FUNDAMENTALS SERIES

Sensitivity and Specificity

Comprehensive Technical Guide: Definitions, General Formulas, and 3×3 Multiclass Evaluation

Source: StatQuest with Josh Starmer — Machine Learning Fundamentals

1. Introduction and Fundamental Concepts

When evaluating machine learning classification models, overall accuracy alone often fails to reveal how a model performs across different classes [1, 4]. A model might achieve high accuracy simply by predicting the majority class, while performing poorly on critical minority cases (such as diagnosing a rare disease).

To overcome this limitation, **Sensitivity** and **Specificity** provide detailed, class-specific evaluation metrics derived directly from a **Confusion Matrix** [2, 4].

Core Definitions

- **Sensitivity (True Positive Rate / Recall):** Measures the proportion of **actual positive cases** that are correctly identified by the model [4]. It answers: "Out of all people who actually have the condition, what percentage did the model correctly catch?"
- **Specificity (True Negative Rate):** Measures the proportion of **actual negative cases** that are correctly identified by the model [4]. It answers: "Out of all people who do not have the condition, what percentage did the model correctly identify as healthy?"

2. Mathematical Formulation for Binary Classification (2×2 Matrix)

In binary classification (e.g., predicting **Heart Disease (+)** vs. **No Heart Disease (-)**), the results are organized into a 2×2 matrix where **rows represent predictions** and **columns represent actual true labels** [2, 3].

		Actual Truth (Known Label)	
		Disease (+)	No Disease (-)
ML Prediction	Disease (+)	True Positive (TP) Sick correctly identified as Sick	False Positive (FP) Healthy misclassified as Sick
	No Disease (-)	False Negative (FN) Sick misclassified as Healthy	True Negative (TN) Healthy correctly identified as Healthy

Figure 1: Diagrammatic Illustration of a 2×2 Binary Confusion Matrix.

2.1 General Formulas

$$\text{Sensitivity} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \quad (1)$$

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}} \quad (2)$$

2.2 Binary Classification Case Study: Heart Disease Prediction

Using testing data from clinical patients [5–7]:

Table 1: Side-by-Side Comparison of Logistic Regression vs. Random Forest [5–7].

Model Algorithm	TP / FN	TN / FP	Sensitivity	Specificity
Logistic Regression	139 / 32	112 / 20	$\frac{139}{139+32} = \mathbf{0.81}$ (81%)	$\frac{112}{112+20} = \mathbf{0.85}$ (85%)
Random Forest	142 / 29	110 / 22	$\frac{142}{142+29} = \mathbf{0.83}$ (83%)	$\frac{110}{110+22} = \mathbf{0.83}$ (83%)

Key Takeaway: Model Selection Trade-offs

- **Random Forest** achieves higher Sensitivity (83% vs 81%). Choose Random Forest if **failing to identify sick patients (False Negatives) carries severe health risks** [7, 8].
- **Logistic Regression** achieves higher Specificity (85% vs 83%). Choose Logistic Regression if **misclassifying healthy patients (False Positives) causes unnecessary harm or expensive invasive tests** [7, 8].

3. Multiclass Extension (3×3 Confusion Matrix)

When evaluating classification problems with 3 or more categories, a critical conceptual shift occurs [9]:

Crucial Video Takeaway: No Overall Matrix Metric

For confusion matrices larger than 2×2 , **there is no single Sensitivity or Specificity value for the entire matrix** [9]. Instead, Sensitivity and Specificity must be calculated **separately for each individual category** using a *One-vs-Rest* formulation [9, 14].

3.1 Example 3×3 Matrix: Movie Preference Prediction

Consider predicting a person's favorite movie among three choices: **Troll 2**, **Gore Police**, and **Cool as Ice** [8, 9].

3.2 Category 1: Sensitivity and Specificity for *Troll 2*

To calculate metrics for *Troll 2*, treat *Troll 2* as the Positive class and all other movies (*Gore Police* + *Cool as Ice*) as the Negative class [10, 11]:

- **True Positives (TP):** 12 (Actual Troll 2, predicted Troll 2) [10].
- **False Negatives (FN):** $102 + 93 = 195$ (Actual Troll 2, but predicted Gore Police or Cool as Ice) [10].

Table 2: The 3×3 Confusion Matrix for Movie Preferences [8, 9].

Predicted \ Actual	Troll 2	Gore Police	Cool as Ice	Total Predicted
Troll 2	12 (TP)	112 (FP)	83 (FP)	207
Gore Police	102 (FN)	23 (TP)	92 (FP)	217
Cool as Ice	93 (FN)	77 (FN)	17 (TP)	187
Total Actual	207	212	192	611

- **True Negatives (TN):** $23 + 77 + 92 + 17 = 209$ (Actual Gore Police/Cool as Ice, and NOT predicted Troll 2) [11].
- **False Positives (FP):** $112 + 83 = 195$ (Actual Gore Police/Cool as Ice, but predicted Troll 2) [11].

$$\text{Sensitivity}_{\text{Troll 2}} = \frac{12}{12 + 195} = \frac{12}{207} \approx \mathbf{0.06} \quad (6\%) \quad (3)$$

$$\text{Specificity}_{\text{Troll 2}} = \frac{209}{209 + 195} = \frac{209}{404} \approx \mathbf{0.52} \quad (52\%) \quad (4)$$

3.3 Category 2: Sensitivity and Specificity for *Gore Police*

Treat *Gore Police* as the Positive class [12, 13]:

- **True Positives (TP):** 23 [12].
- **False Negatives (FN):** $102 + 92 = 194$ (Actual Gore Police, predicted Troll 2 or Cool as Ice) [12].
- **True Negatives (TN):** $12 + 93 + 83 + 17 = 205$ [13].
- **False Positives (FP):** $112 + 77 = 189$ [13].

$$\text{Sensitivity}_{\text{Gore Police}} = \frac{23}{23 + 194} = \frac{23}{217} \approx \mathbf{0.11} \quad (11\%) \quad (5)$$

$$\text{Specificity}_{\text{Gore Police}} = \frac{205}{205 + 189} = \frac{205}{394} \approx \mathbf{0.52} \quad (52\%) \quad (6)$$

3.4 Category 3: Sensitivity and Specificity for *Cool as Ice*

Treat *Cool as Ice* as the Positive class [14]:

- **True Positives (TP):** 17 [14].
- **False Negatives (FN):** $83 + 92 = 175$ [14].
- **True Negatives (TN):** $12 + 102 + 112 + 23 = 249$ [14].
- **False Positives (FP):** $93 + 77 = 170$ [14].

$$\text{Sensitivity}_{\text{Cool as Ice}} = \frac{17}{17 + 175} = \frac{17}{192} \approx \mathbf{0.09} \quad (9\%) \quad (7)$$

$$\text{Specificity}_{\text{Cool as Ice}} = \frac{249}{249 + 170} = \frac{249}{419} \approx \mathbf{0.59} \quad (59\%) \quad (8)$$

4. Summary of Key Takeaways

1. **Sensitivity** measures positive detection rate ($\frac{TP}{TP+FN}$); **Specificity** measures negative detection rate ($\frac{TN}{TN+FP}$) [4, 15].
2. In clinical and high-stakes machine learning applications, model selection depends on whether preventing False Negatives or False Positives is more vital [8, 15].
3. For an $N \times N$ confusion matrix, there are N distinct pairs of Sensitivity and Specificity values [9, 14].